

H. Treasure Hunt

time limit per test: 1 second memory limit per test: 256 megabytes

Fares is trapped in the past during the [Treasure Hunt Contest](#).



To bring him back to the present, the students must collect at least x time crystals to open the time portal.

There are n puzzles. The i -th puzzle gives a_i time crystals, but its crystals are collected only if the puzzle is solved.

Given two arrays, the crystals array and the solved array.

Determine whether the students collected enough crystals to rescue Fares.

Input

The first line contains two integers n and x ($1 \leq n \leq 10^5$, $1 \leq x \leq 10^9$), the number of puzzles and the minimum number of crystals needed.

next line contains n integers $a_1, a_2, \dots, a_n$ ($1 \leq a_i \leq 1000$) representing the crystals array.

next line contains n integers $s_1, s_2, \dots, s_n$ ($0 \leq s_i \leq 1$) representing the the solved array.

Output

Print YES if the total number of collected crystals is at least x . Otherwise, print NO.

Example

Input

```
5 3
3 9 1 8 7
0 0 0 0 0
```

Output

```
NO
```